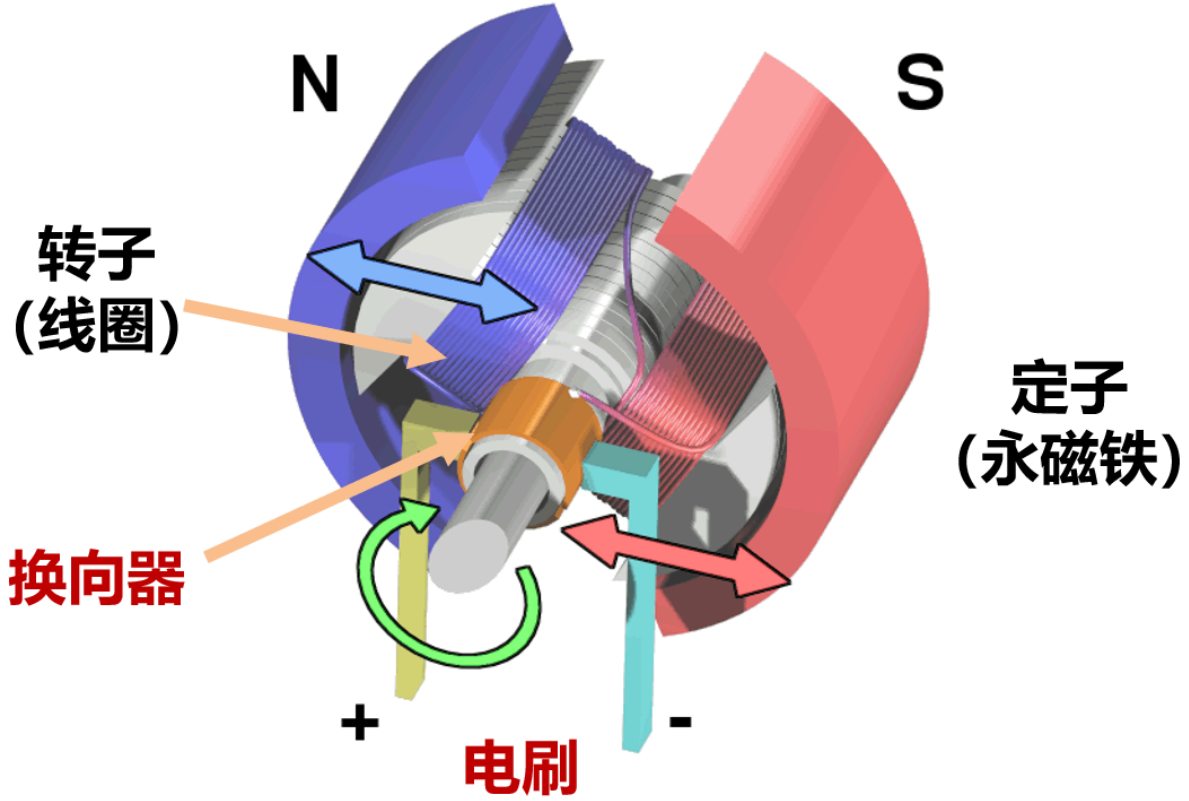
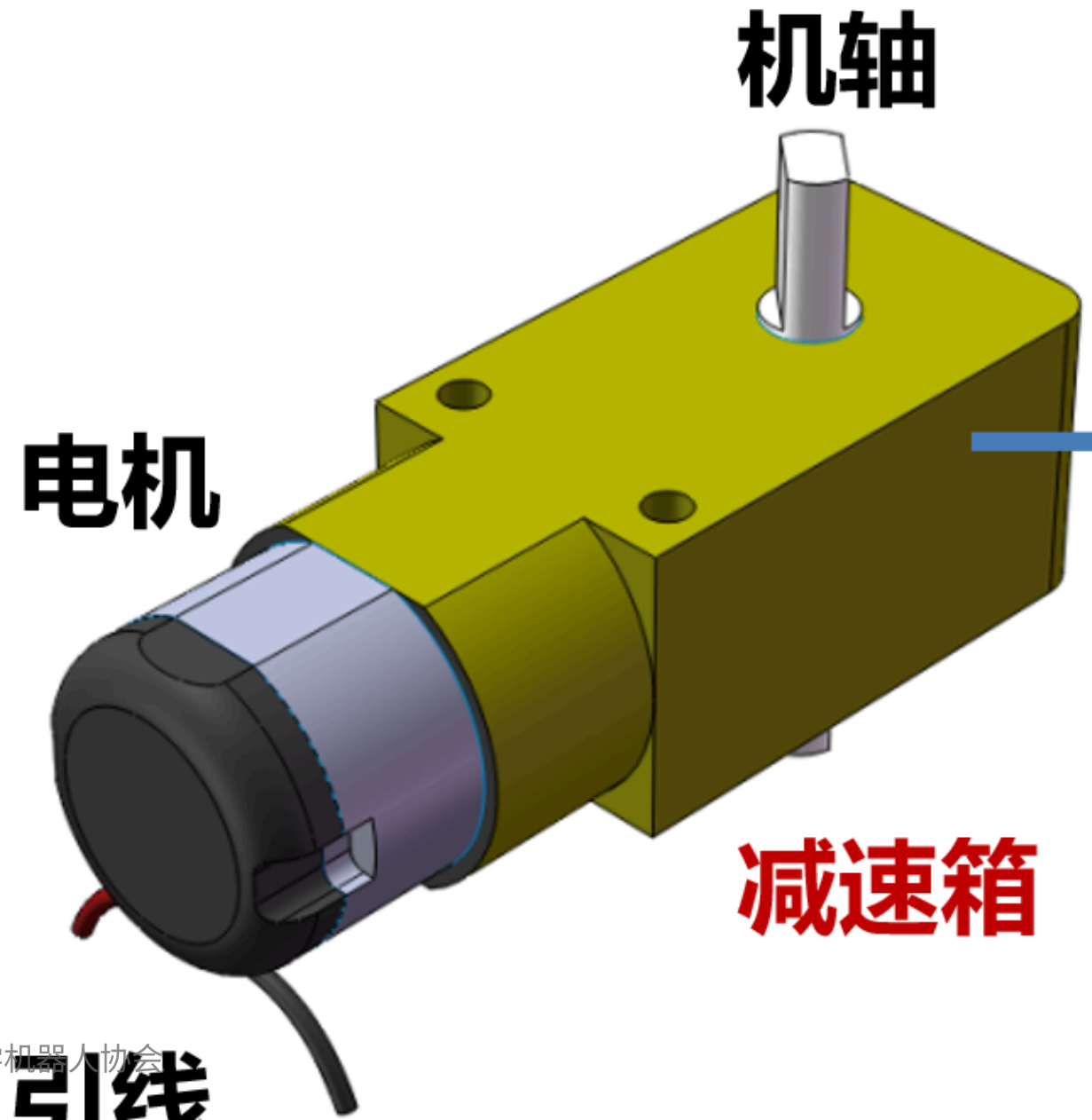
电机驱动原理

有刷直流电机

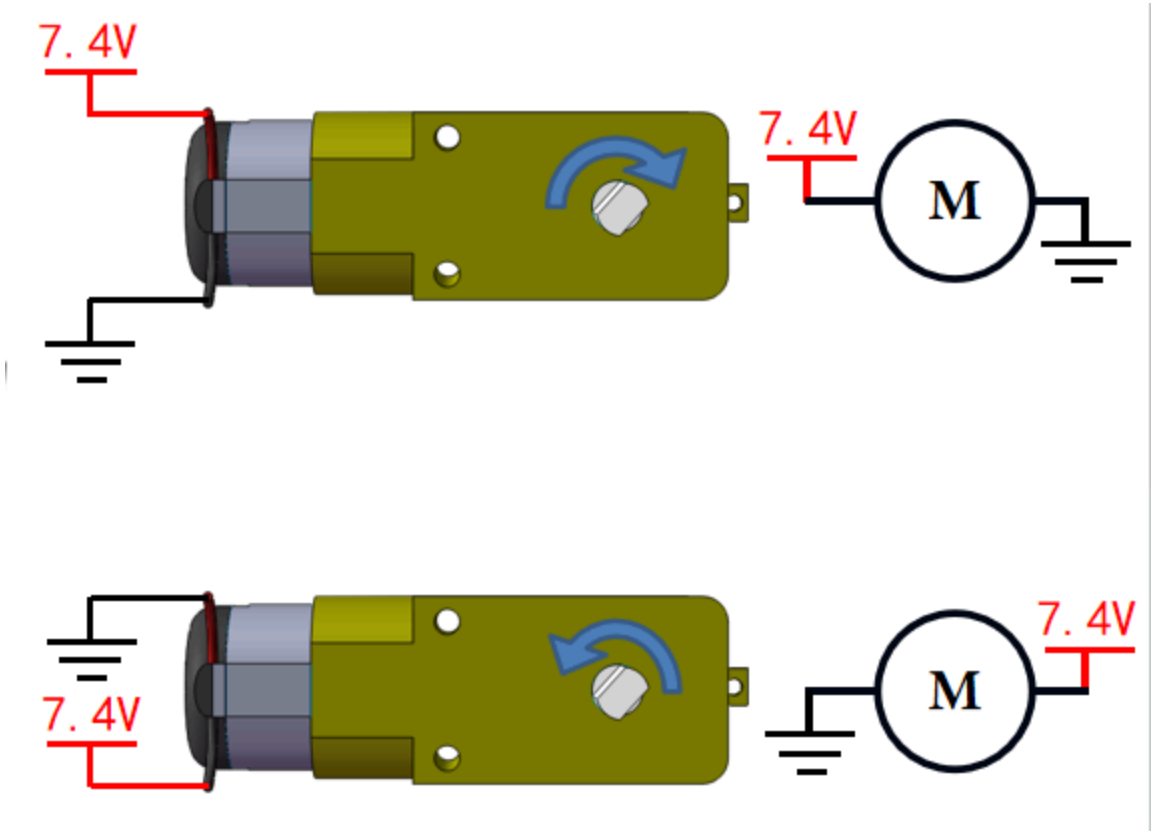
Brushed DC Motor





以齿轮的组合
减小转速，加大扭矩

直流电机的正反转



单片机对电机的控制

在一个电路中实现电机的正反转——半导体器件、H桥电路

实现对转速的调控——PWM调制

拓展：PN结、二极管、BJT和FET

- PN结：半导体、微电子器件的基础
- 二极管：简单的PN结应用器件 单向导通
- 三极管（BJT）：双极型晶体管；分为NPN、PNP；可起开关作用
- 场效应管（FET）：集成电路的核心器件；原理、应用都类似三极管

半导体（Si/Ge）

N型半导体

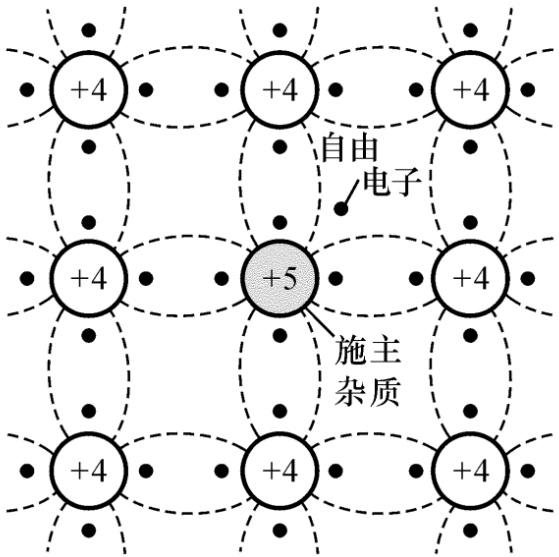
- 掺少量5价元素（磷、砷等）；
- 内含很多自由电子。

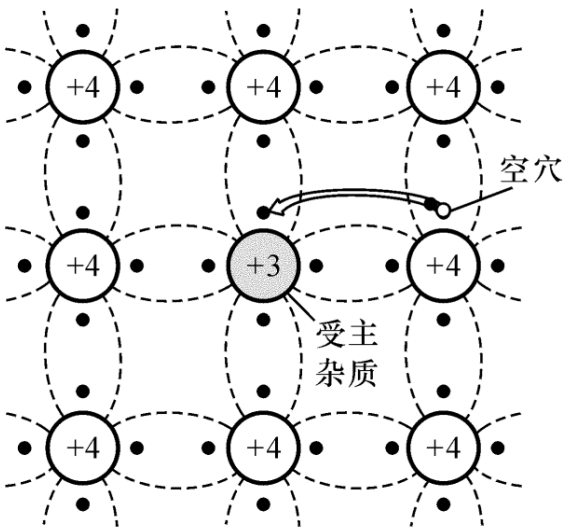
P型半导体

- 掺少量3价元素（硼、镓等）；
- 内含很多空穴（正电荷）。

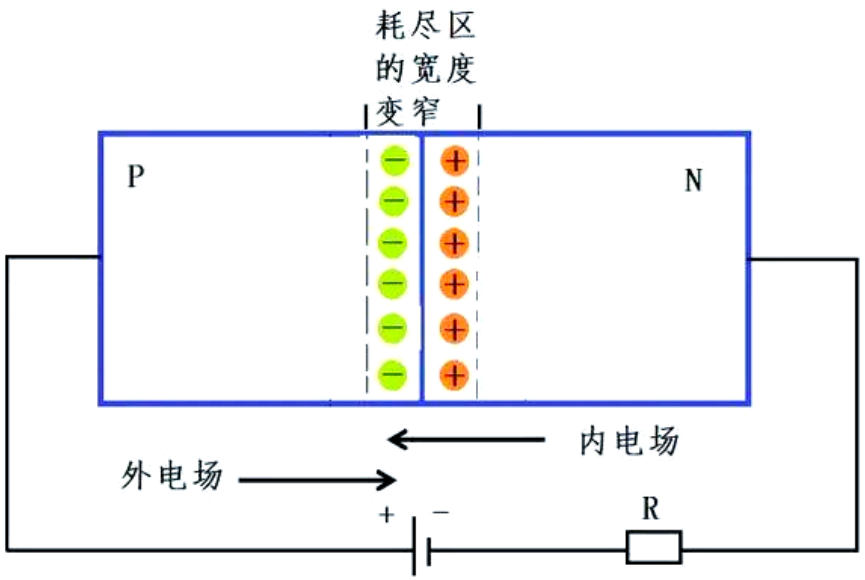
PN结

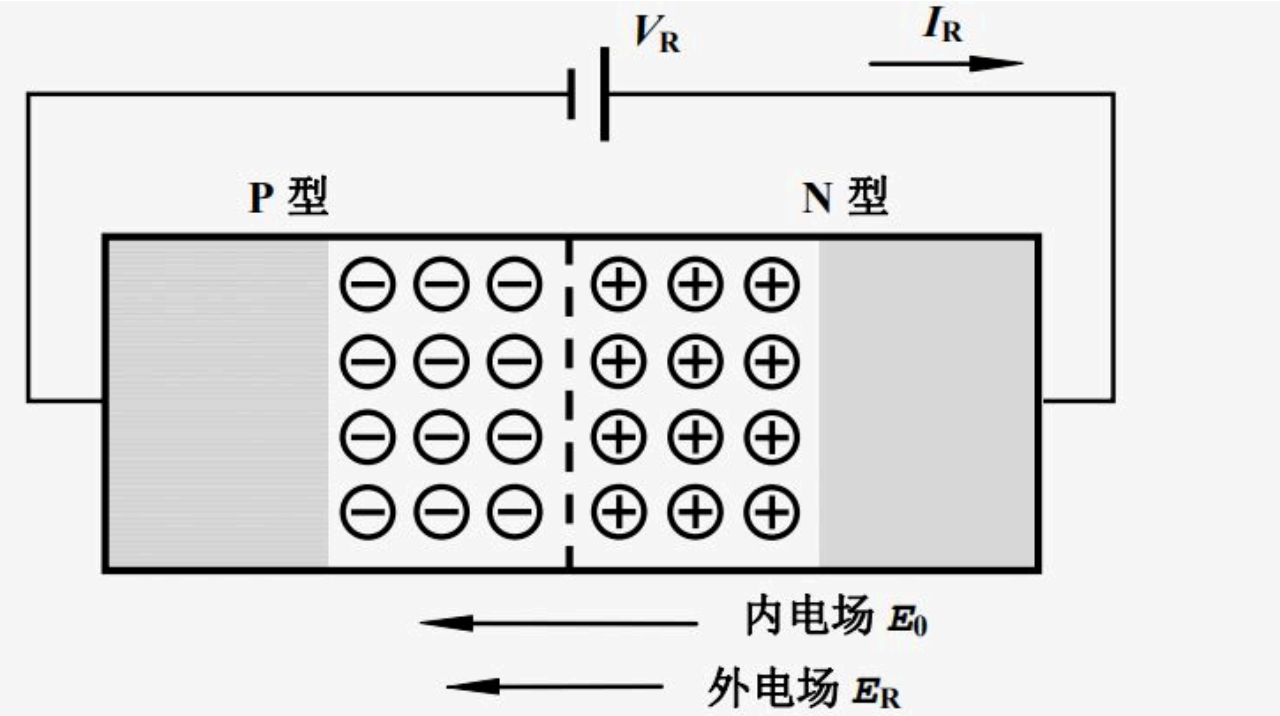
PN结：两种不同类型的半导体在交界面附近形成的空间电荷层。





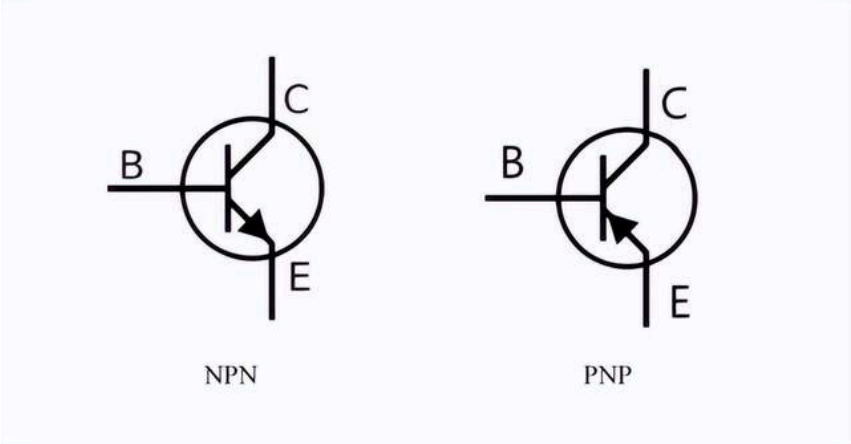
PN结与二极管

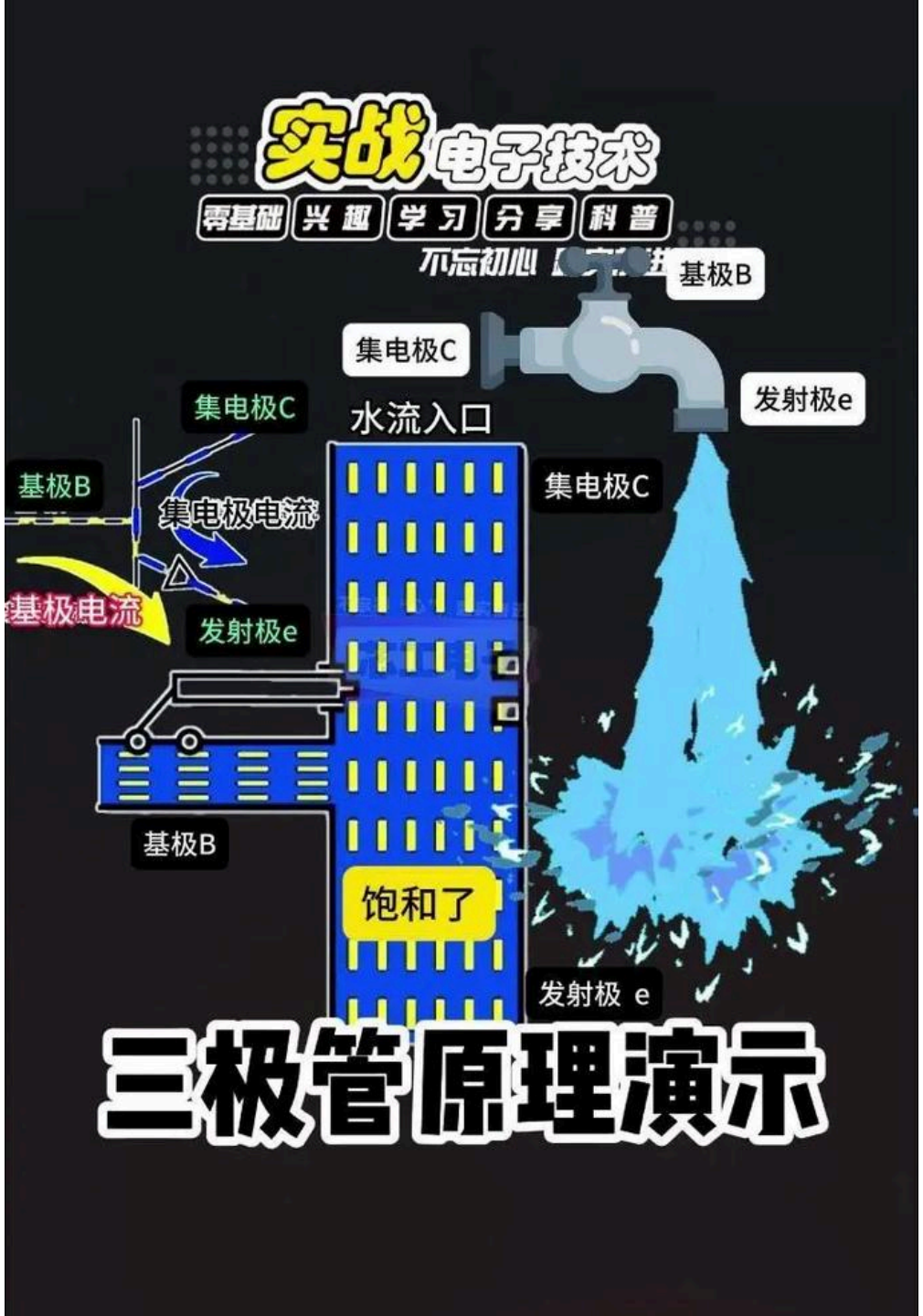




三极管

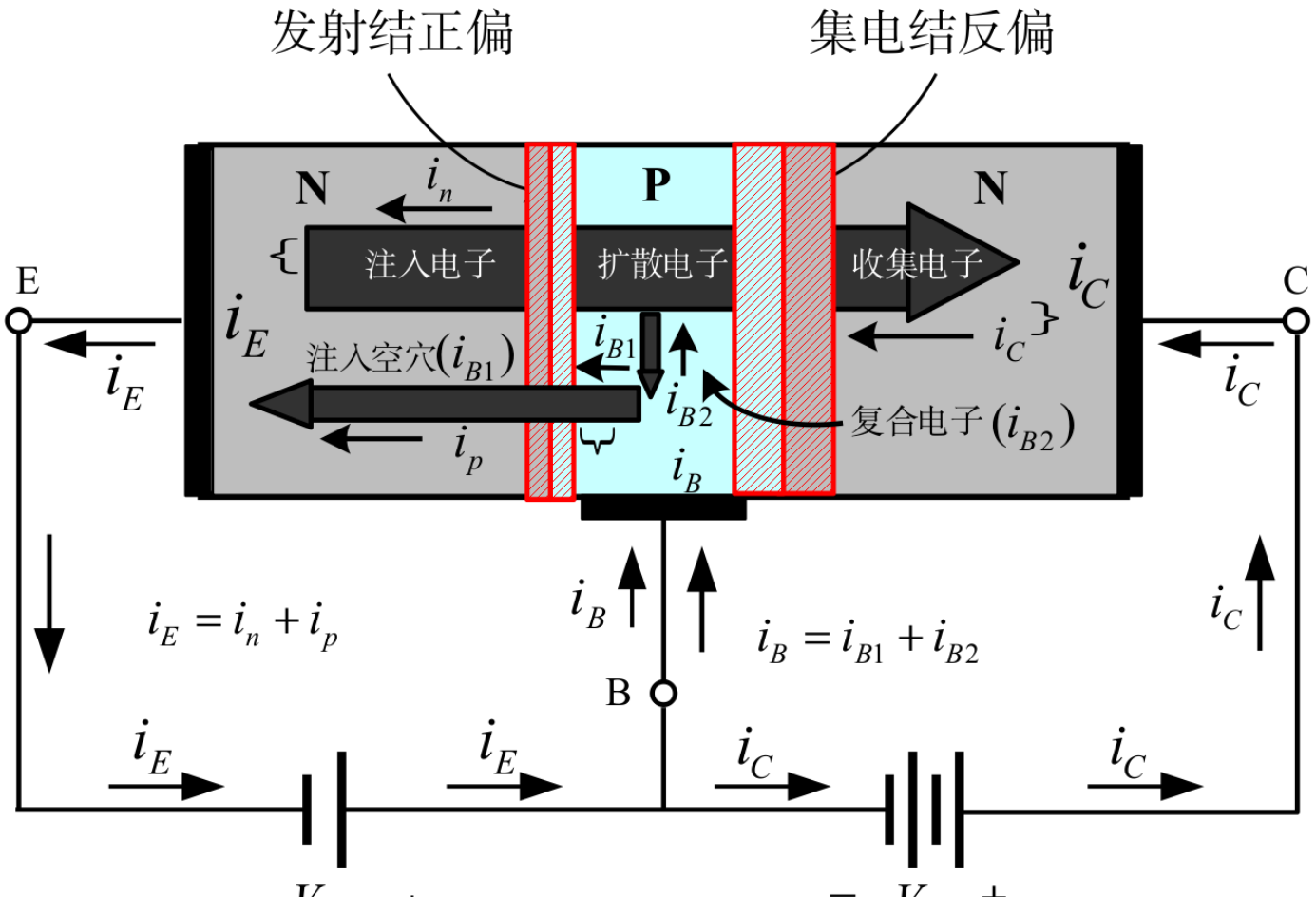
- 全称：双极型晶体管（Bipolar Junction Transistor）
- 分类：NPN、PNP
- 核心原理：像水龙头的阀门一样，控制流过的电流大小

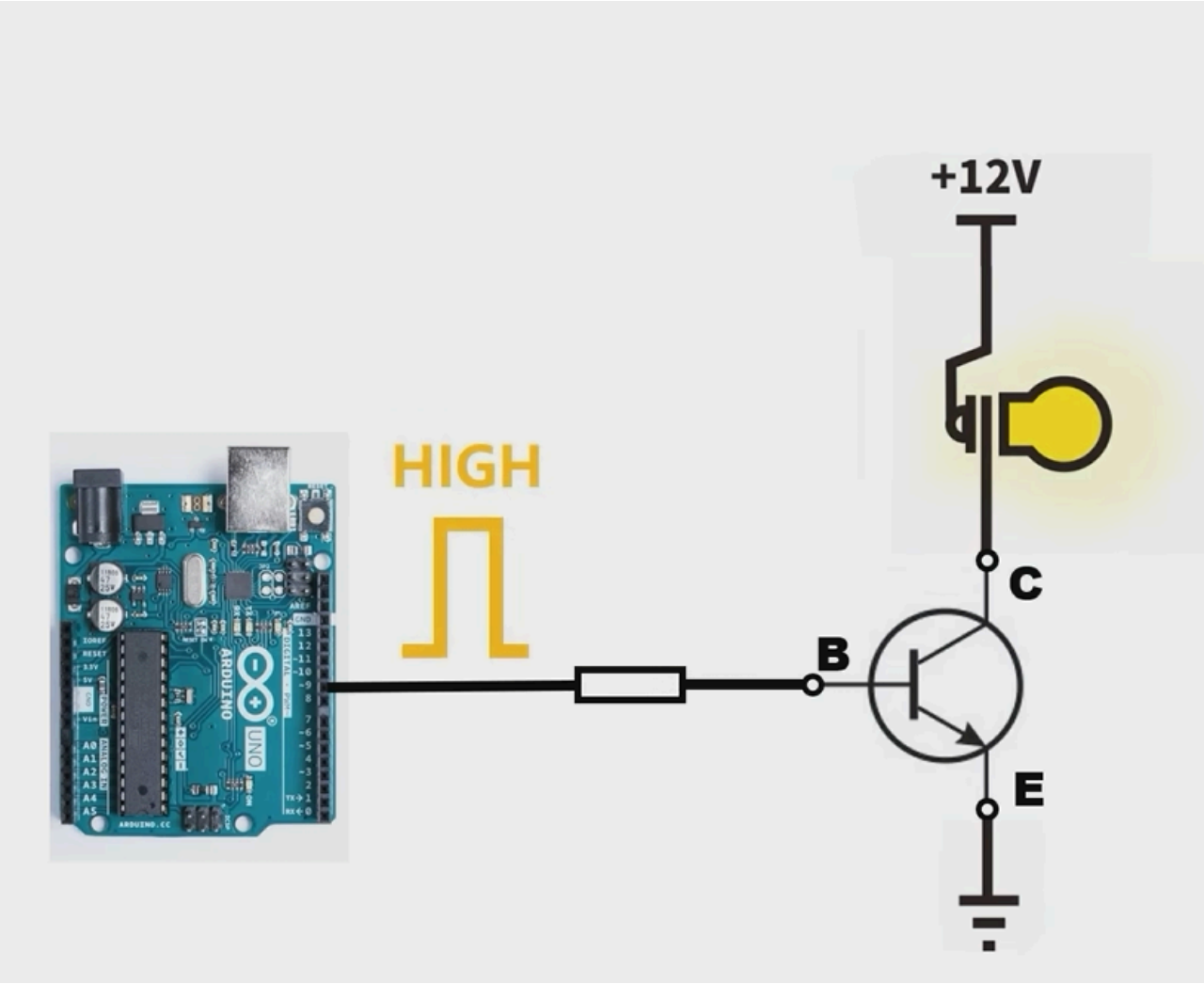


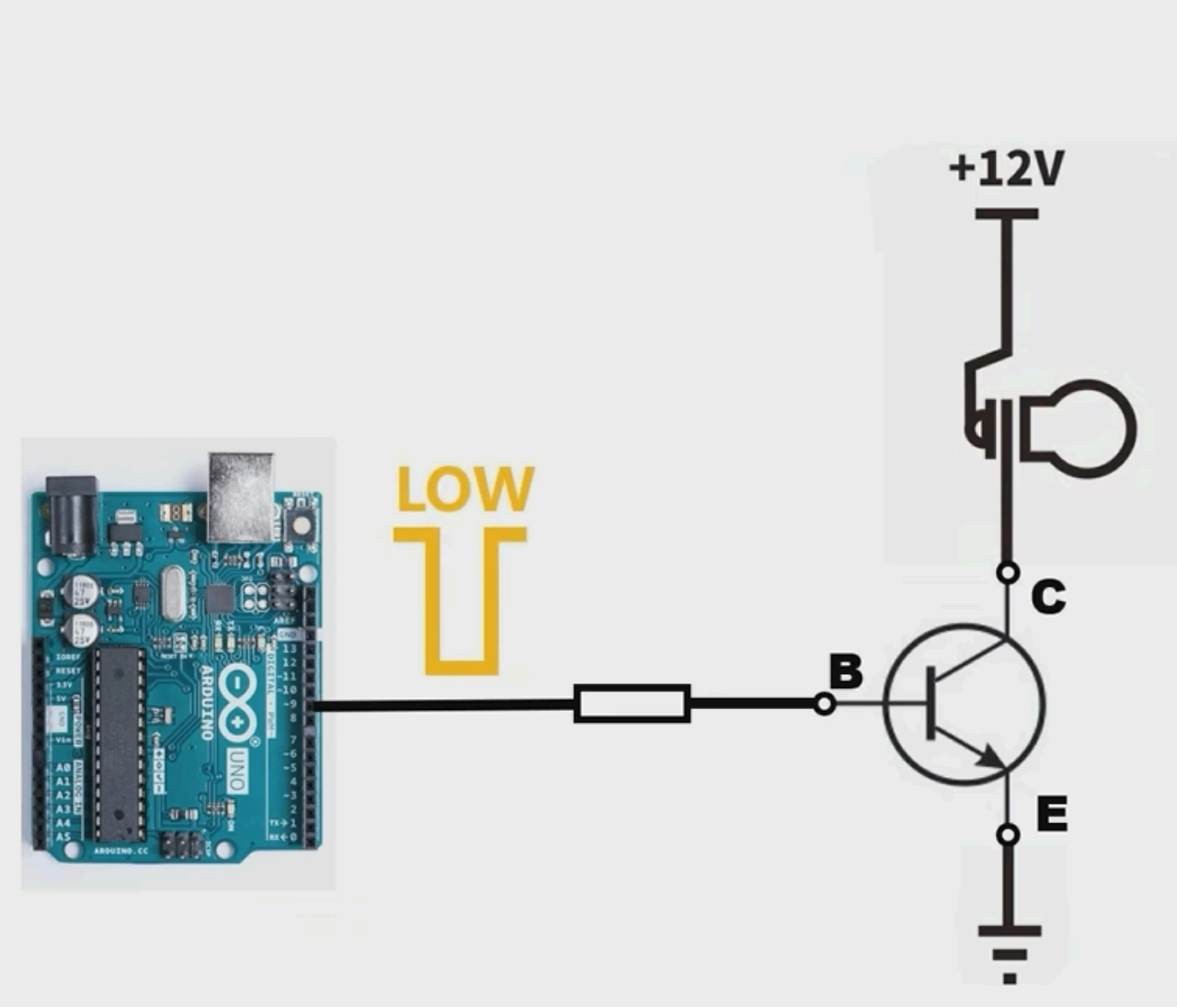


三极管原理——电流控制元件

（本图借鉴了电子电路基础chap7的ppt）



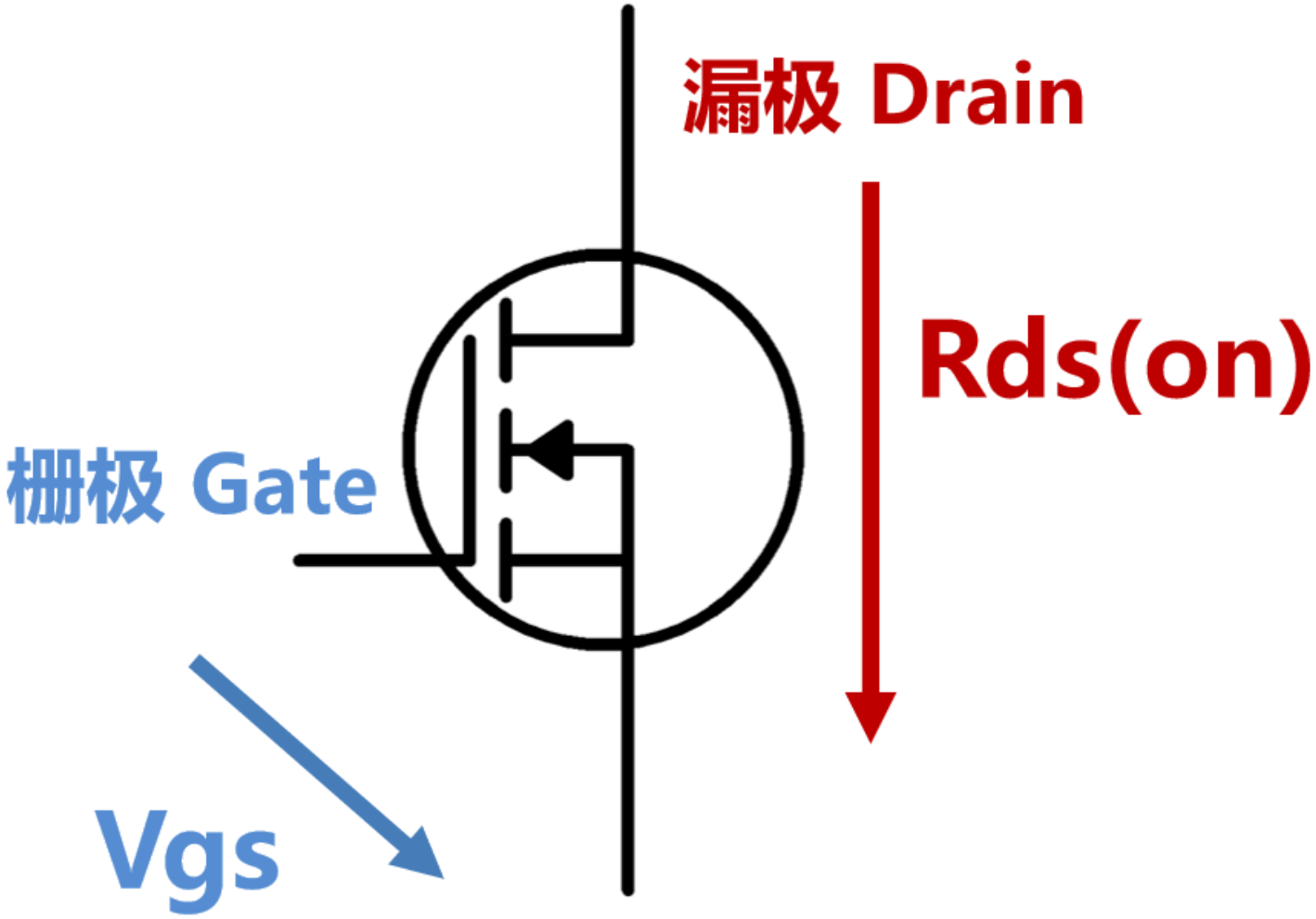

+12V
HIGH
C
B
E


+12V
LOW
C
B
E

场效应管（Field Effect Transistor）

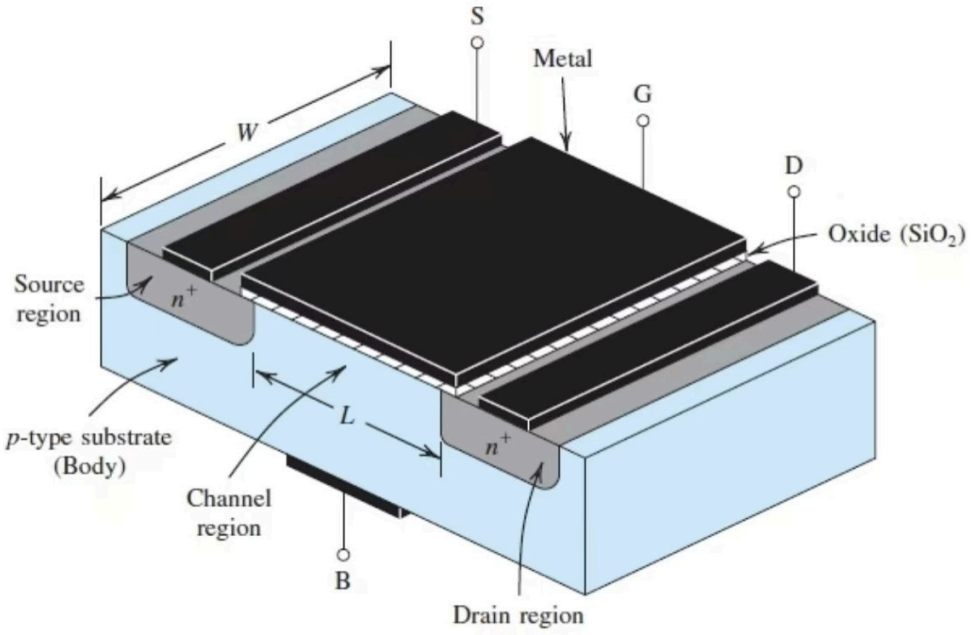
- 集成电路的核心器件；种类样式繁多（MOSFET、FinFET等）
- 以MOSFET为例，下分增强型/耗尽型 NMOS/PMOS
- NMOS可以看作NPN型三极管，PMOS可以看作PNP型三极管
- CMOS:互补型（complementary）金属氧化物晶体管

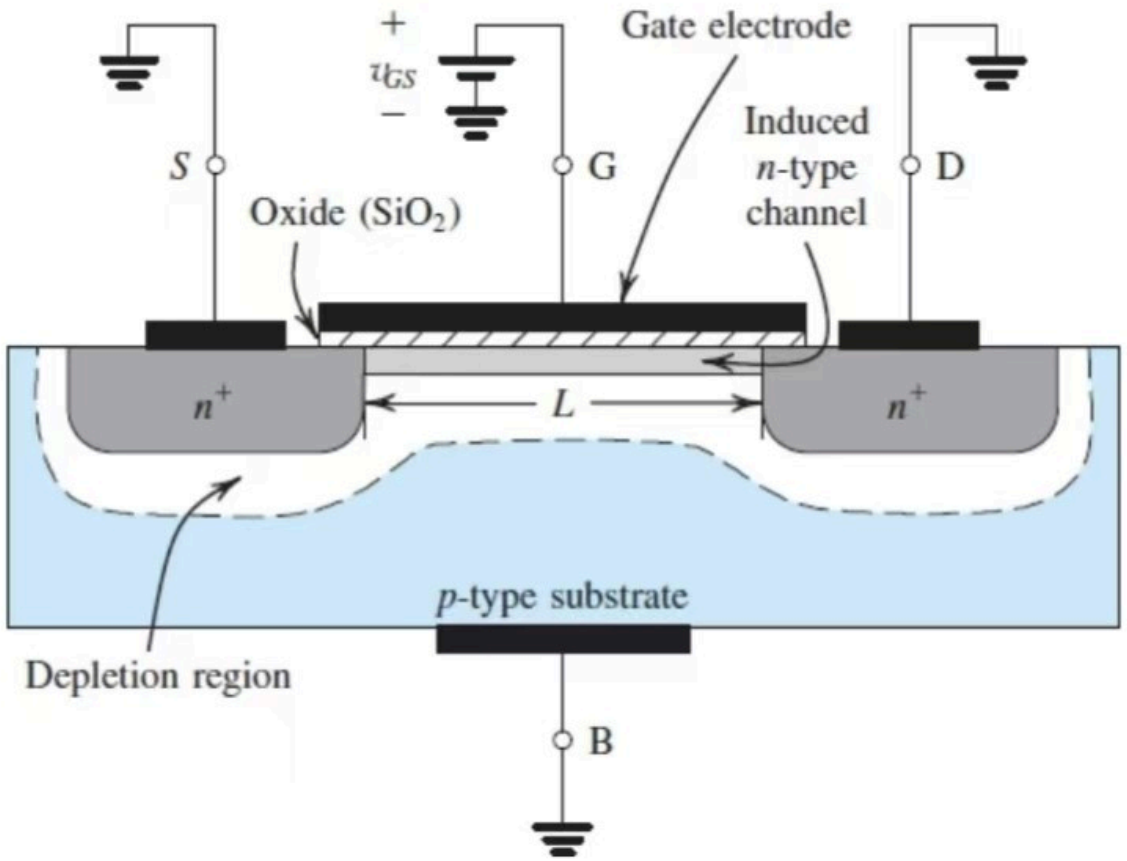
N沟道增强型场效应管 NMOS

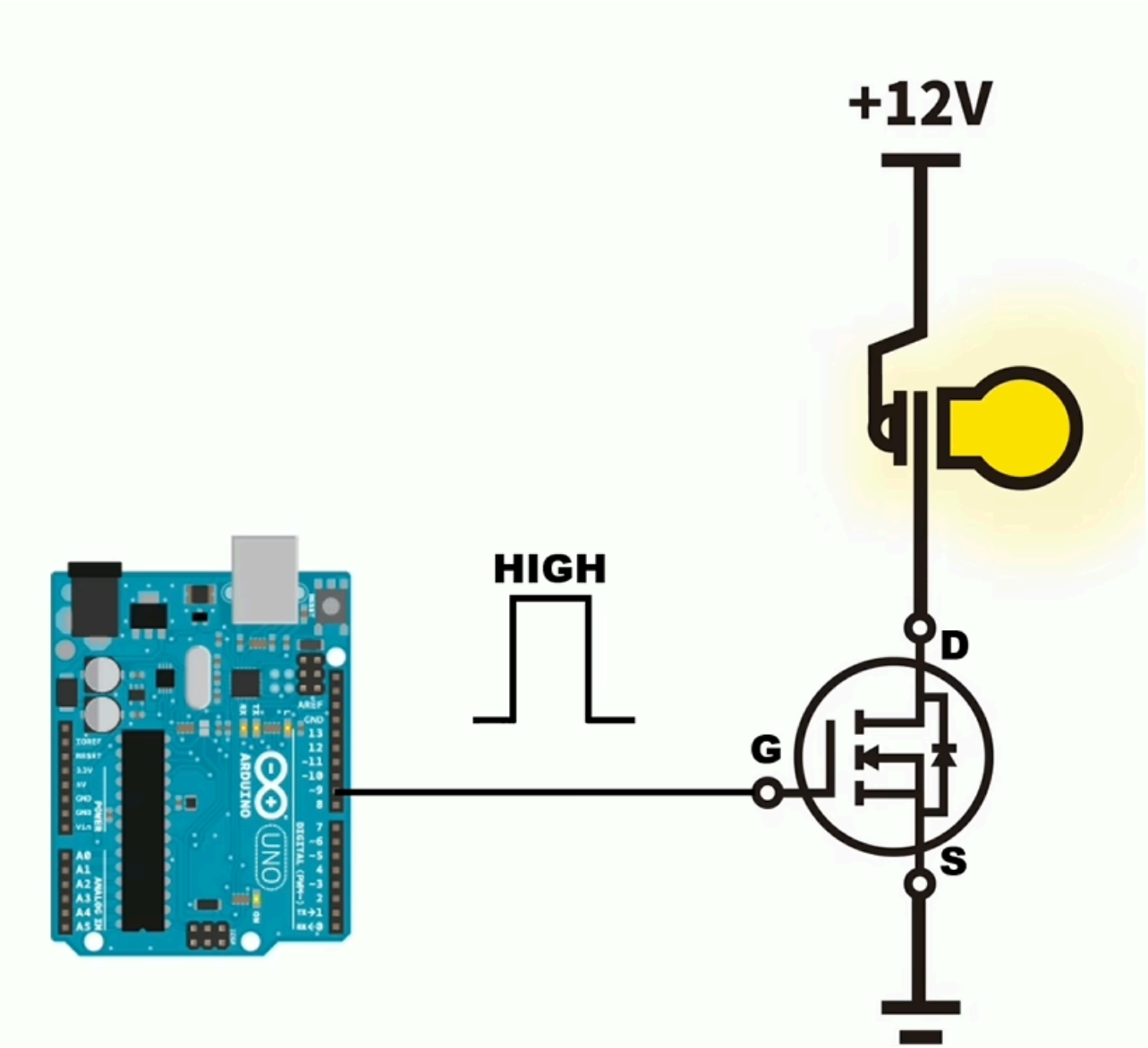


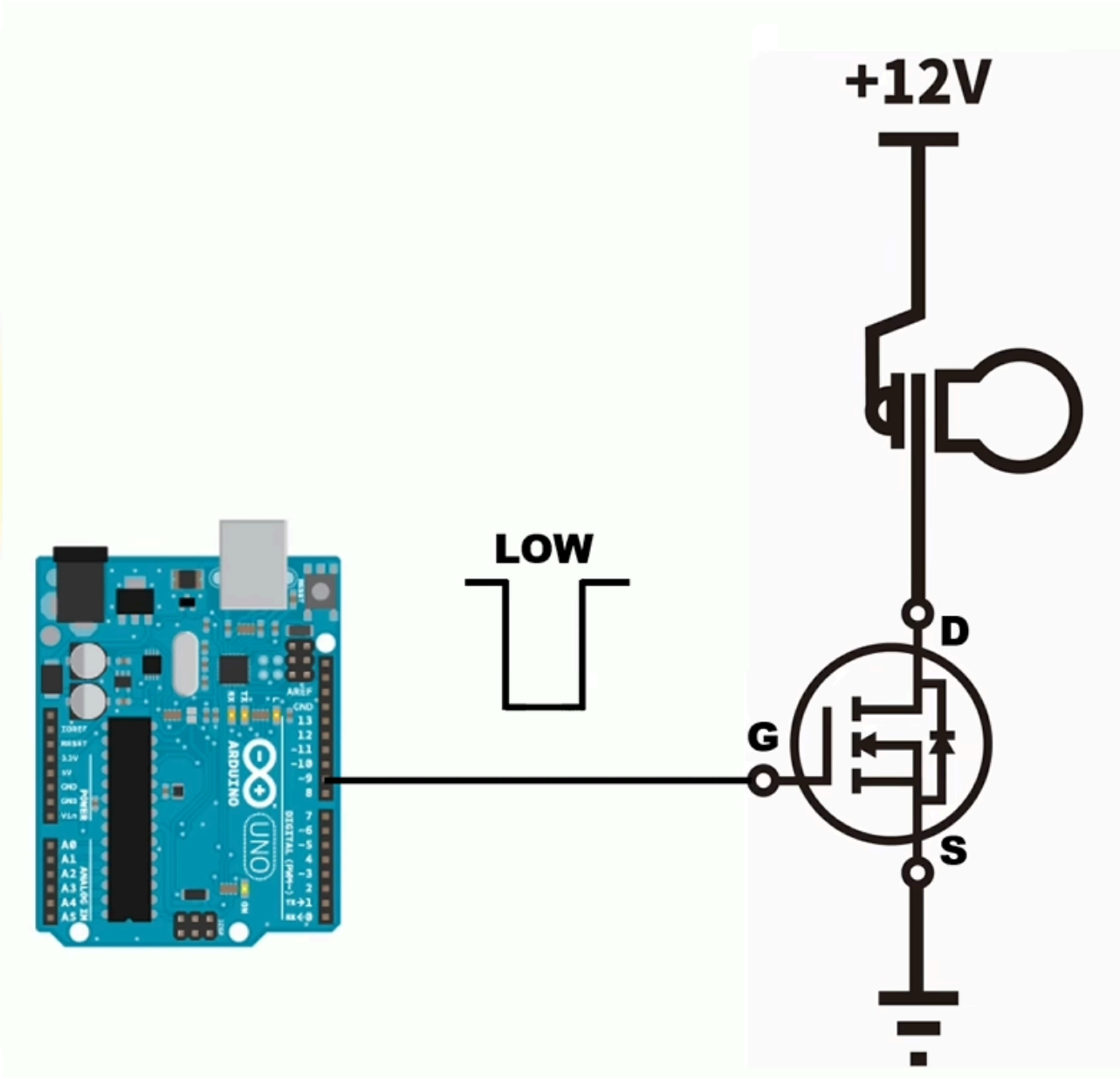
源极 Source

在内部与衬底相连


S
Metal
G
W
D
Oxide (SiO_2)
Source region
n^+
p-type substrate (Body)
L
Channel region
n^+
B
Drain region

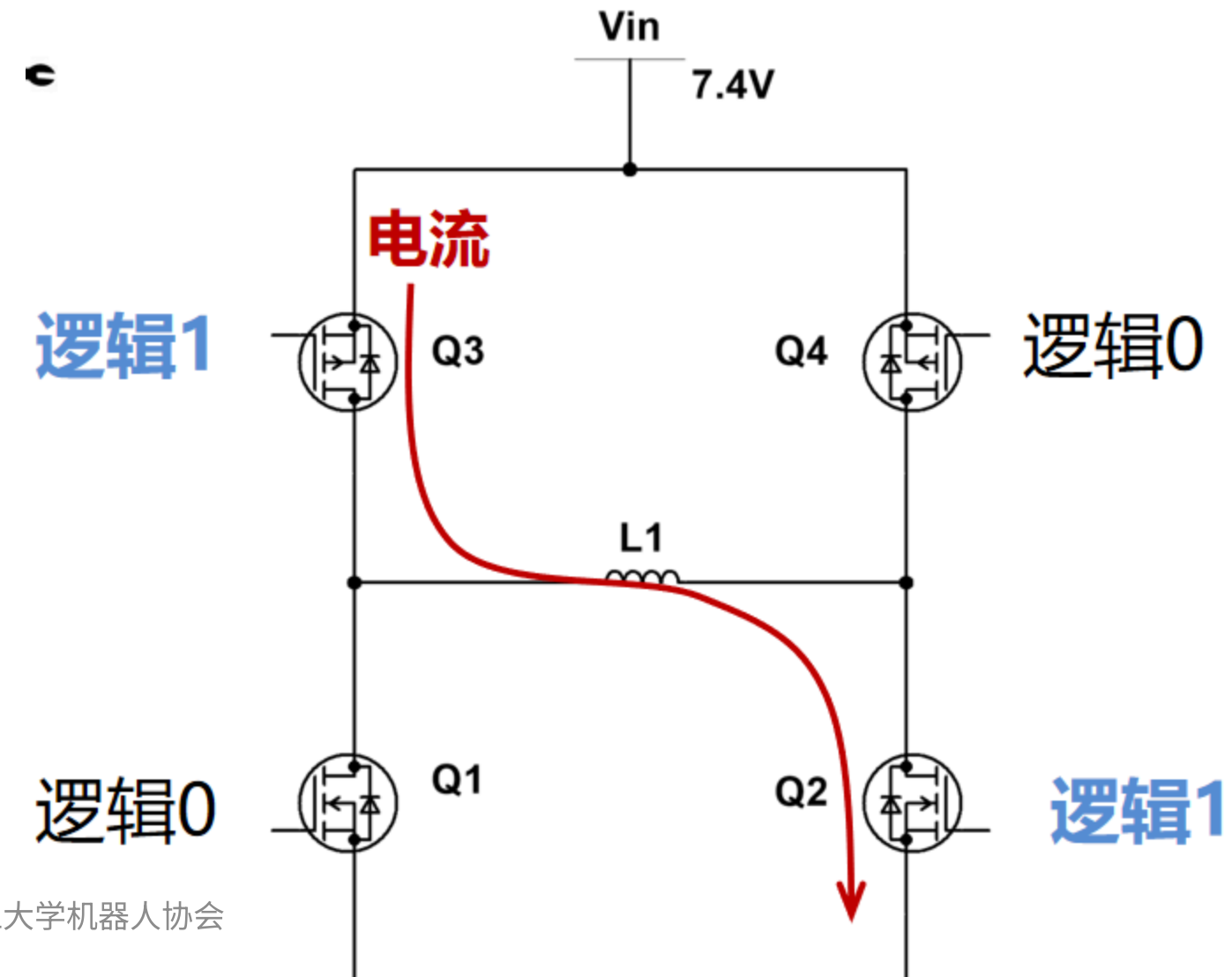

+
v_{GS}
−
Gate electrode
S
G
D
Oxide (SiO_2)
Induced n-type channel
n^+
L
n^+
p-type substrate
Depletion region
B

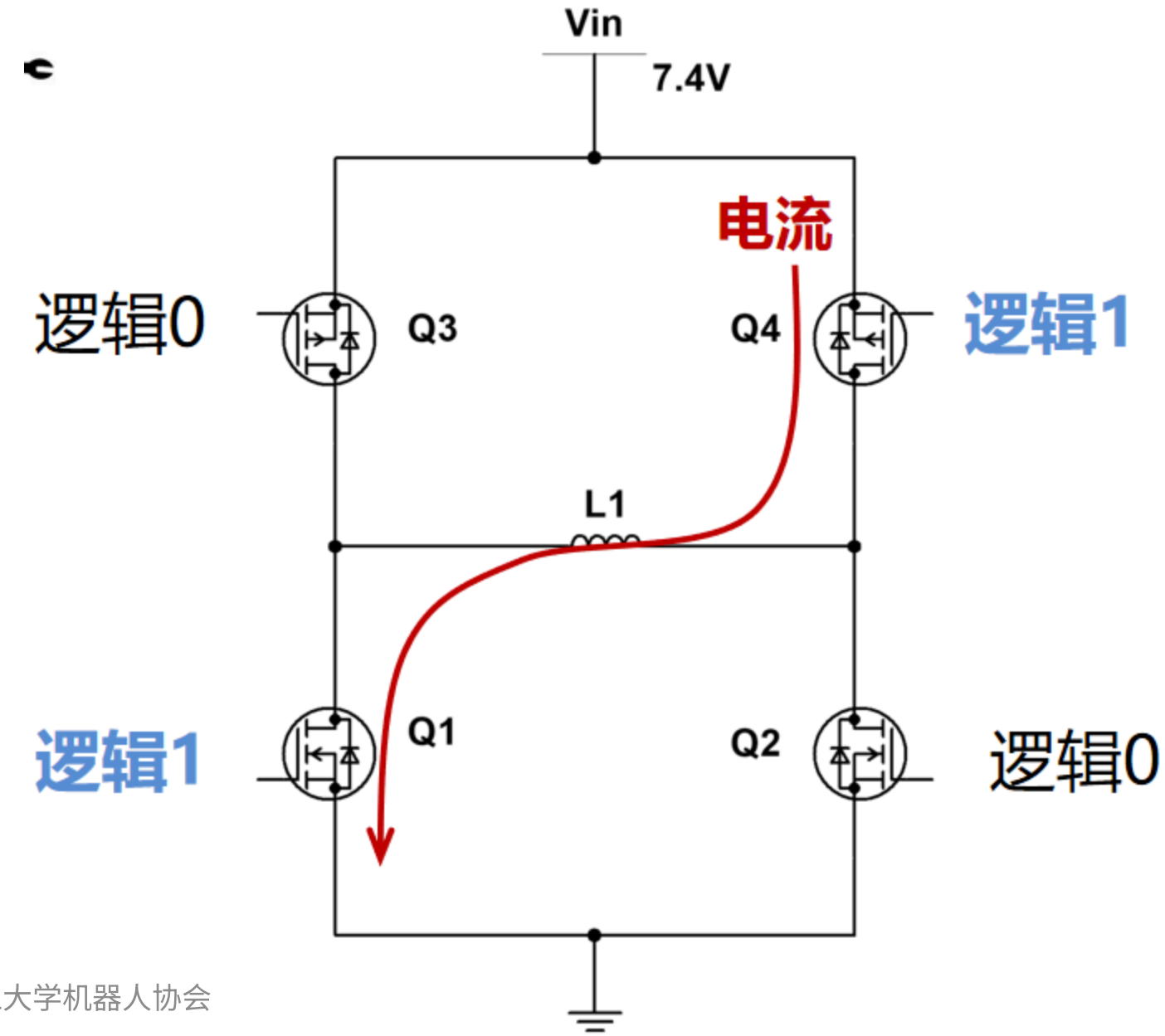

+12V
HIGH
D
G
S

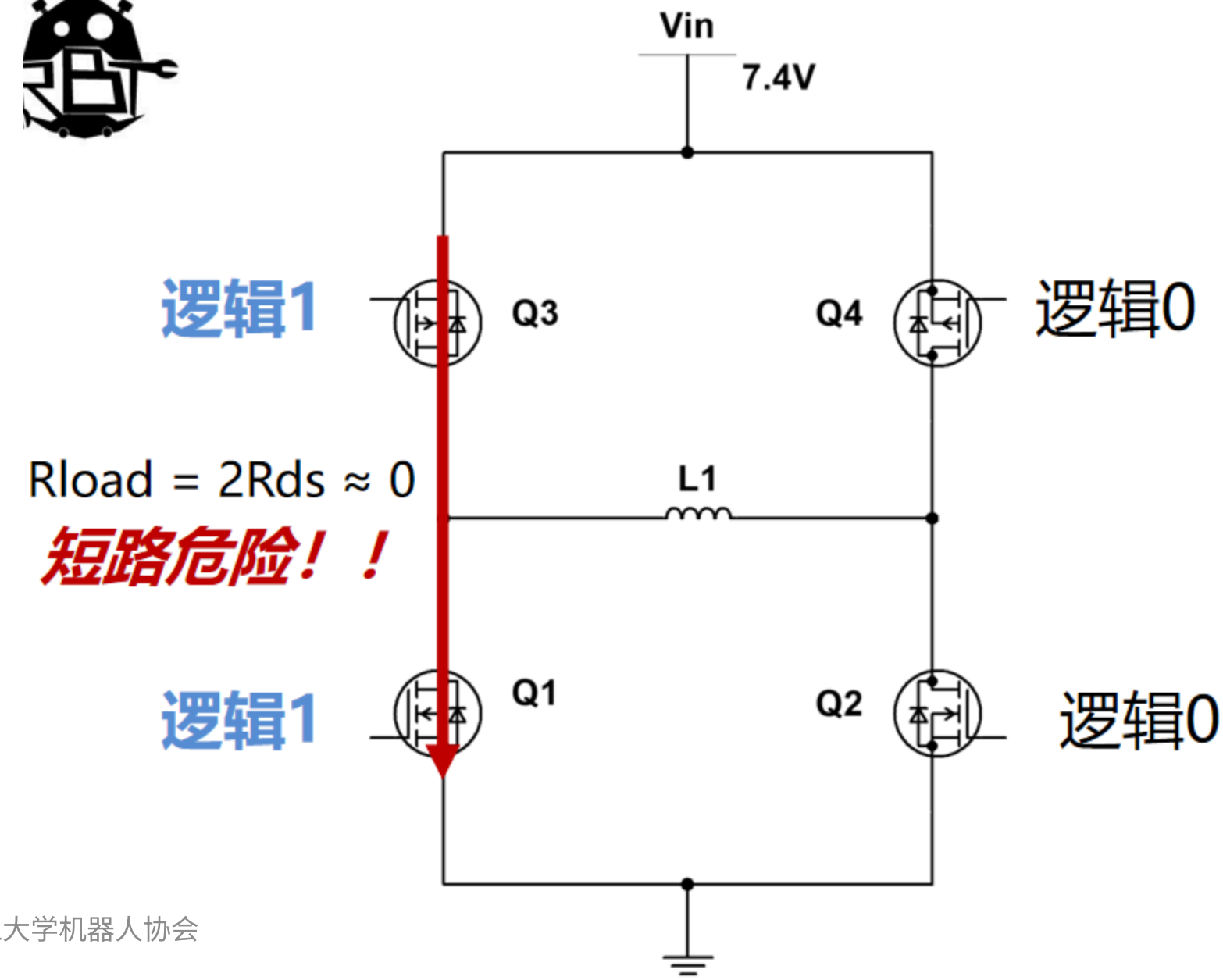

+12V
LOW
D
G
S

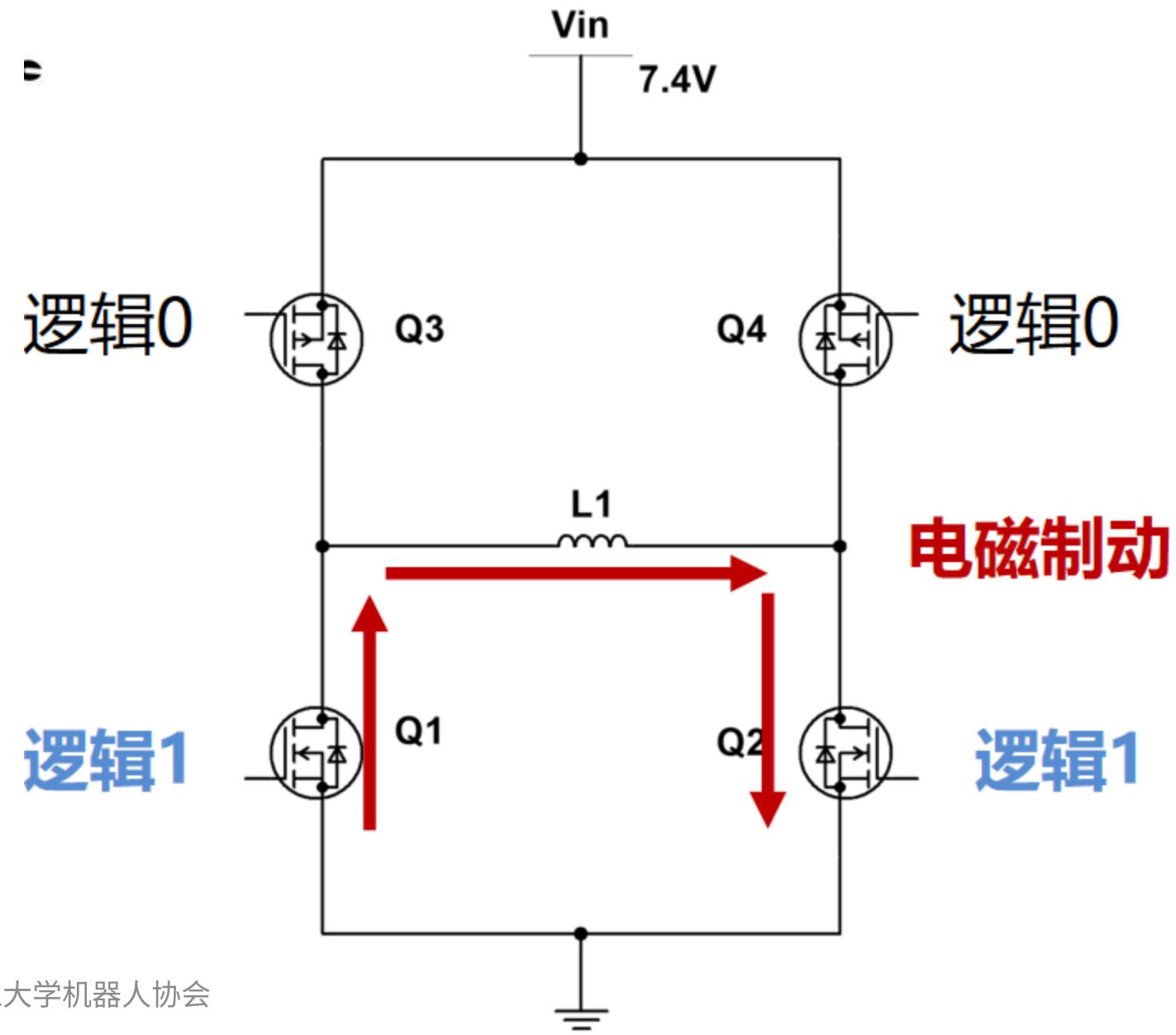
总而言之，你在这里可以把它当成开关/电磁继电器/水龙头/whatsoever

回归主线




Vin
7.4V
电流
逻辑0
Q3
Q4
逻辑1
L1
逻辑1
Q1
Q2
逻辑0


Vin
7.4V
逻辑1
Q3
Q4
逻辑0
Rload = 2Rds ≈ 0
L1
短路危险！！
逻辑1
Q1
Q2
逻辑0


Vin
7.4V
逻辑0
Q3
Q4
逻辑0
L1
电磁制动
逻辑1
Q1
Q2
逻辑1

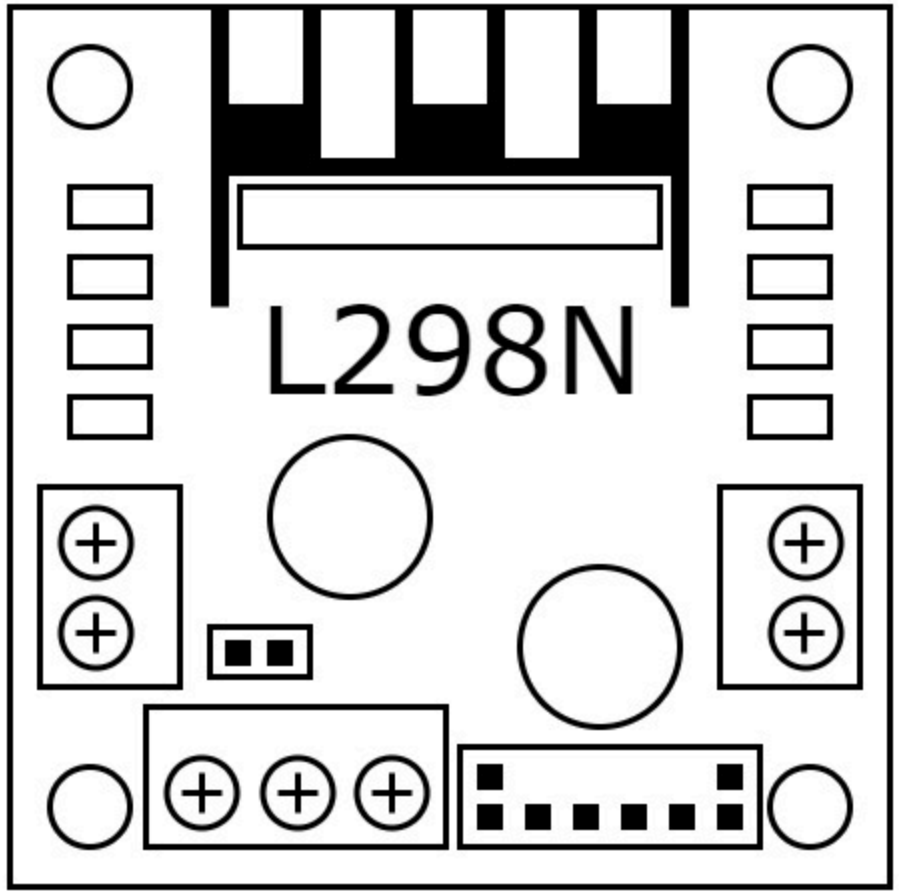
控制转速

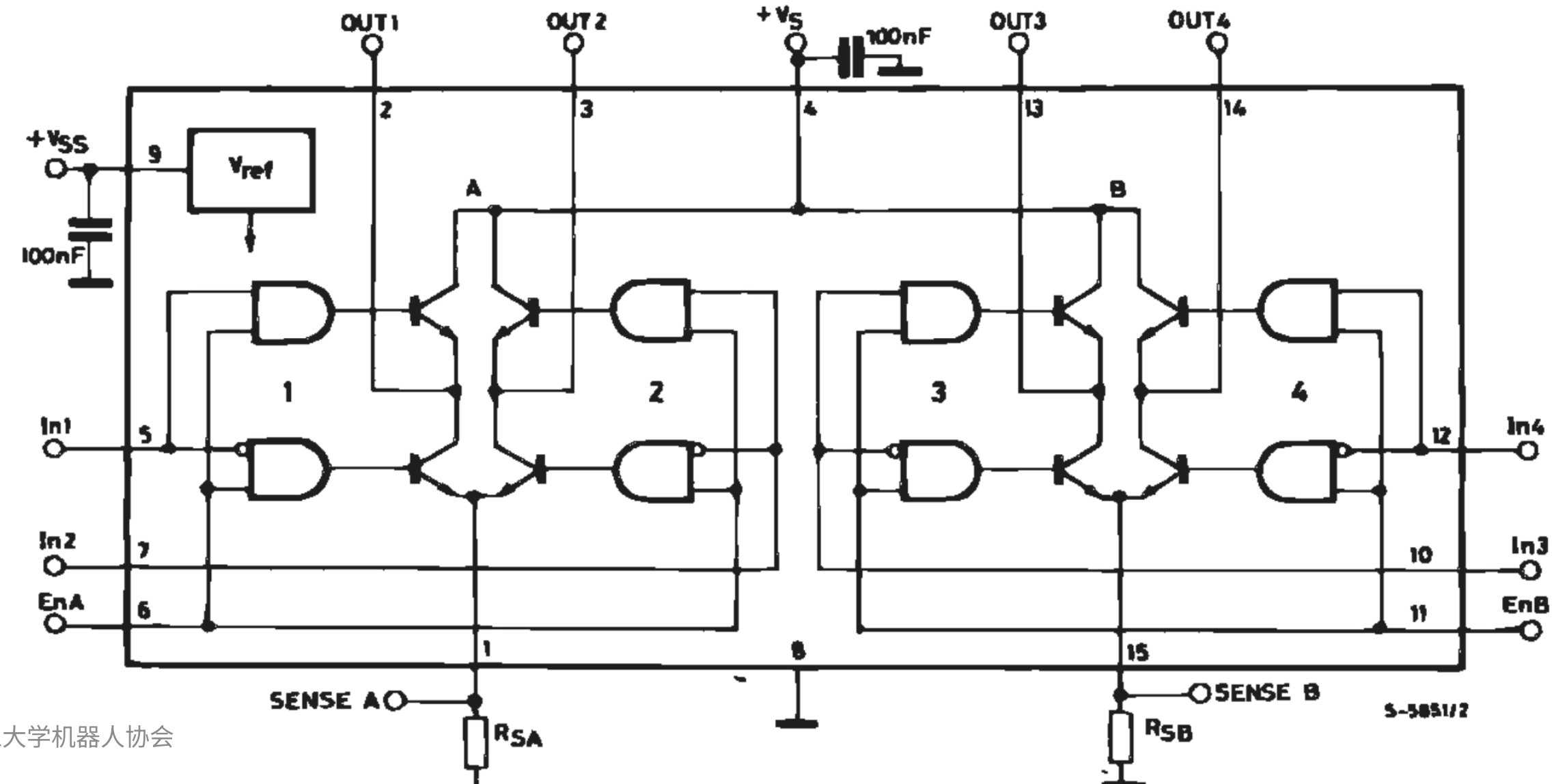
电机是一个惯性系统。控制电机的转速，可以通过给电机输入高频率的脉冲电压信号，对电机的效果可以等效为冲量相等的直流电。

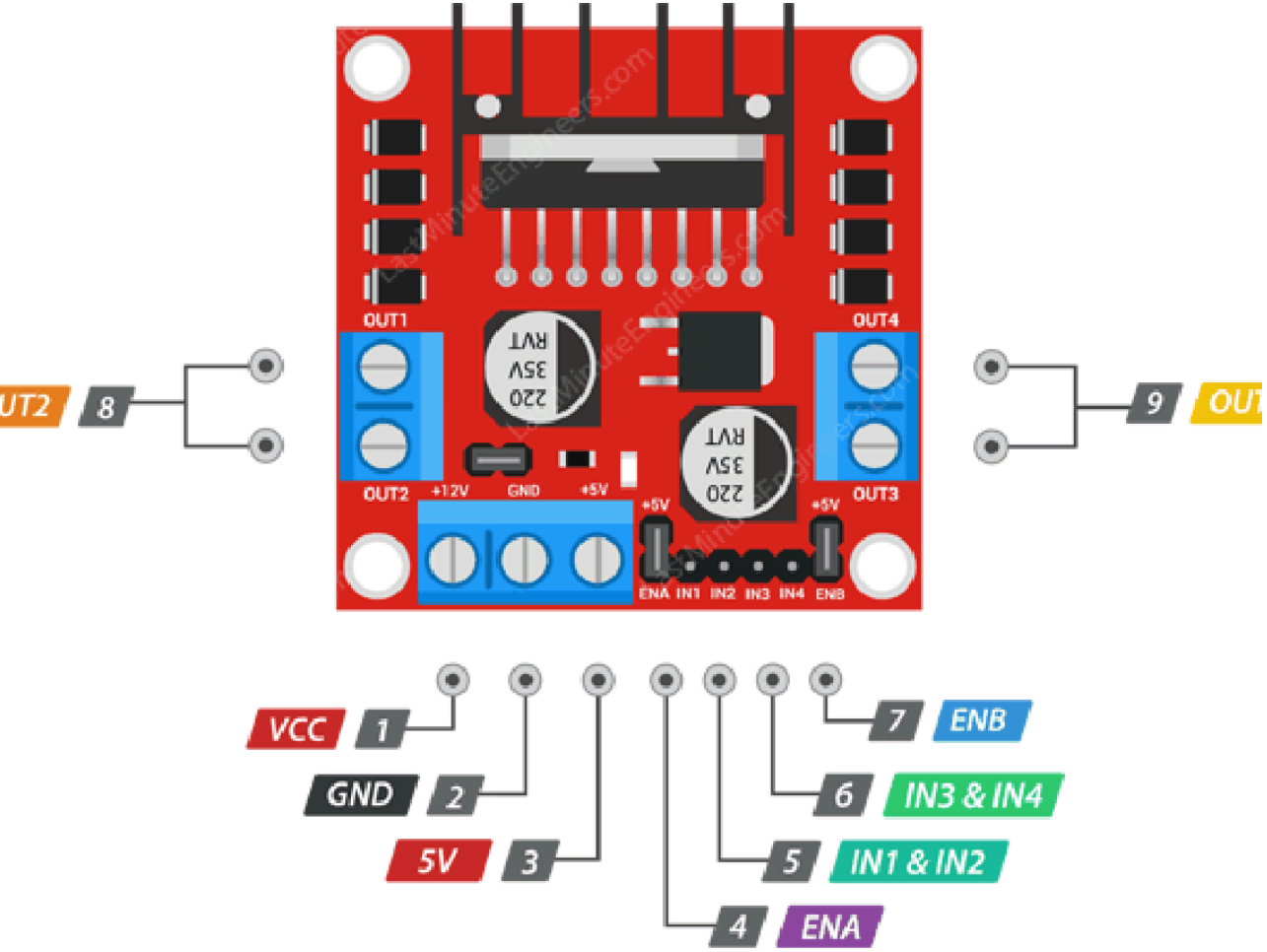
脉冲宽度调制（PWM）：不同的占空比等效为不同的直流信号。

Arduino的PWM输出： analogWrite(pin, value)；

电机驱动——L298N

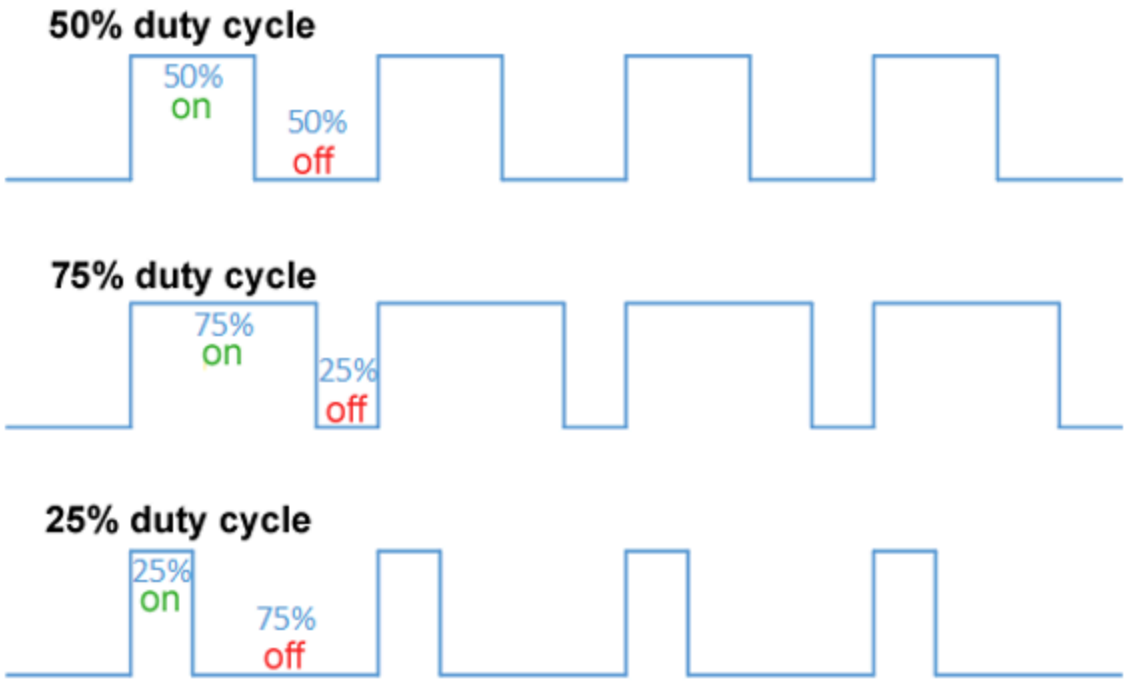



OUT1
OUT2
+VS
100nF
OUT3
OUT4
2
3
4
13
14
+VSS
9
Vref
A
B
100nF
1
2
3
4
In1
5
12
In4
In2
7
10
In3
EnA
6
11
EnB
1
8
15
SENSE A
SENSE B
RSA
RSB
S-5851/2



EnA	In1	In2	状态
0	-	-	停止
1	1	0	正转
1	0	1	反转
1	1	1	急停
1	0	0	滑行

ENA =


50% duty cycle
50%
on
50%
off
75% duty cycle
75%
on
25%
off
25% duty cycle
25%
on
75%
off

占空比太低，电机不转?

绕组电流

$$u_L(t) = L\frac{di_L}{dt}$$

$$i_L(t) = \frac{1}{L}\int_0^t u_L dt$$

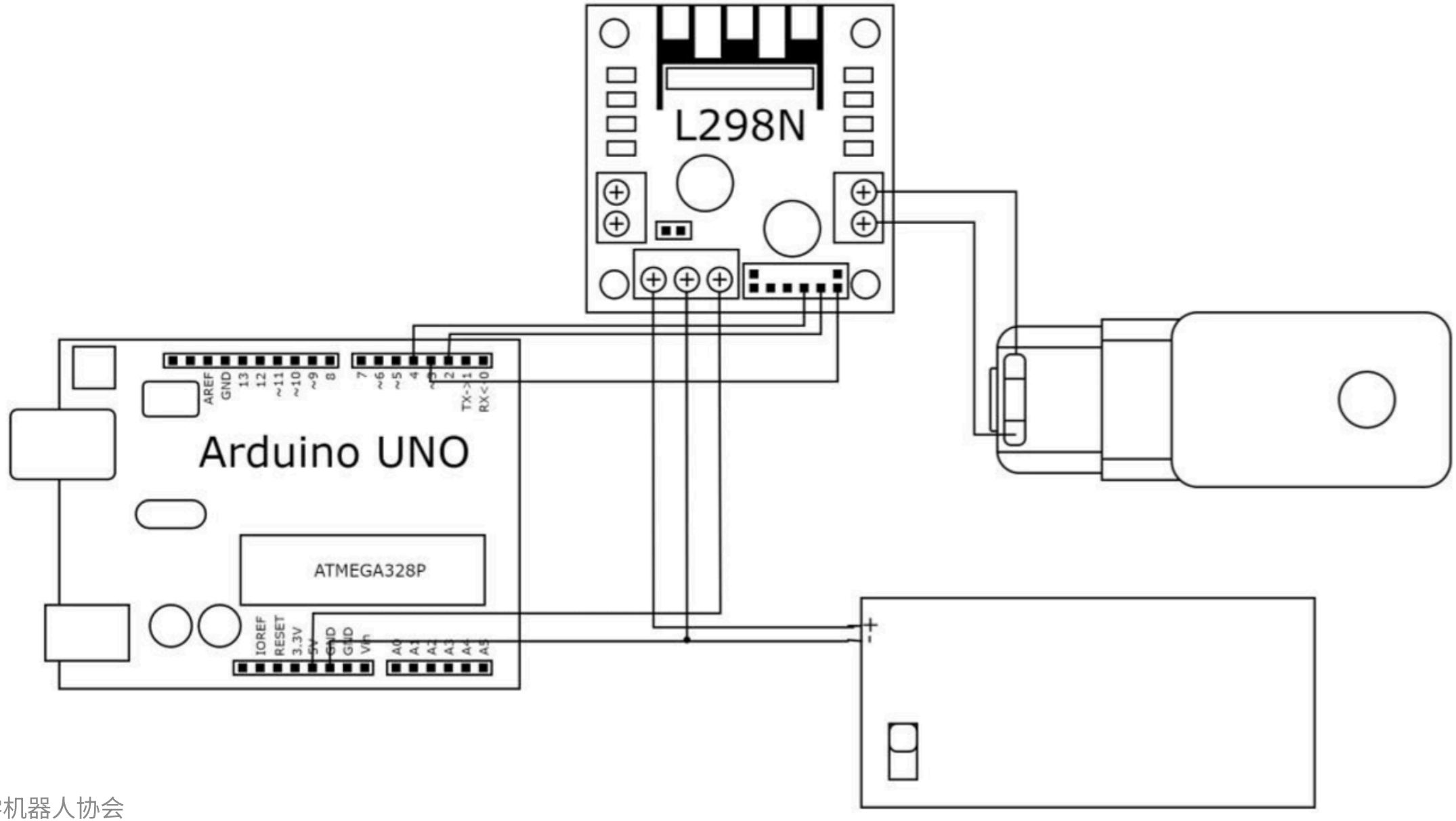
+

磁力矩

$$M = NBI\sin\theta$$

↓

绕组电感充电时间太短，电流过小
不足以克服阻力


L298N
Arduino UNO
ATMEGA328P
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX->1
RX<0
IOREF
RESET
3.3V
5V
GND
GND
Vin
A0
A1
A2
A3
A4
A5
+
-